

Practical No. 09

Aim: Develop Decision Tree Classification Model for given dataset and use it to classify new sample

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# Step 1: Import necessary libraries
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt

# Step 2: Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# Step 3: Split into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Step 4: Train Decision Tree Classifier
clf = DecisionTreeClassifier(criterion='gini', max_depth=3, random_state=42)
clf.fit(X_train, y_train)

# Step 5: Evaluate model
y_pred = clf.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy:.2f}")

# Step 6: Classify new sample
new_sample = [[5.1, 3.5, 1.4, 0.2]] # Example: one flower's features
predicted_class = clf.predict(new_sample)
print(f"Predicted class for the new sample: {iris.target_names[predicted_class[0]]}")

# Step 7: Visualize the tree
plt.figure(figsize=(10, 6))
plot_tree(
    clf,
    filled=True,
    feature_names=iris.feature_names,
    class_names=iris.target_names
)
plt.title("Decision Tree for Iris Dataset")
plt.show()
```

Model Accuracy: 1.00
Predicted class for the new sample: setosa

Decision Tree for Iris Dataset

